

PROBLEM STATEMENT #61 | Sustainability & Green Tech

Household Carbon Footprint & Target Tracker

1. Problem Context & Overview

A green living analytics tool where households log monthly utility consumption (kWh electricity, gas, commute kilometers) to compute \$CO_2\$ equivalents and track reduction milestones.

Target Stakeholders / Actors: *Resident, Sustainability Coach*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (\$CO_2e\$) emissions.

Sample Acceptance Criteria: Pass: Calculated emissions match EPA conversion factor formulas; Fail: Zero or negative emission values generated.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The annual carbon reduction milestone chart must render interactive historical comparison graphs in under 200 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.